

Abhishek Vincent

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Professional Summary

Platform engineer with hands-on experience building security infrastructure, microVM orchestration systems, and ephemeral lab environments. Shipped production systems handling 50+ concurrent isolated VM sessions, won national awards, and disclosed critical vulnerabilities responsibly. Strong in DevOps, Kubernetes, Firecracker, and platform tooling.

Education

Yenepoya University, Bangalore Bachelor of Computer Applications (BCA) in Cybersecurity, Ethical Hacking, Digital Forensics Aggregate: 87.9%	2023 – 2026
St Aloysius HSS, Kollam, Kerala Class 12 – Computer Science (State Board) 78%	2021 – 2023
St Aloysius HSS, Kollam, Kerala Class 10 (State Board) 90%	2020 – 2021

Experience

Platform Engineer (Contract) – ZecurX Cybersecurity Pvt. Ltd	Feb 2026 – Present
• Architected and deployed Nexus MicroVM – a Firecracker-based ephemeral VM platform serving isolated per-session lab environments for CTF and cybersecurity training • Designed ZFS copy-on-write storage layer enabling sub-second rootfs cloning across 5 OS images (Kali, Ubuntu, Debian, Fedora, BlackArch), reducing storage overhead by 95% vs. naive full-copy approach • Built Nexus Conductor (Go daemon) handling full VM lifecycle: provisioning, TAP networking, Jailer sandboxing, WebSocket terminal proxying, warm pool management, and session expiry enforcement • Implemented warm pool strategy maintaining 10 pre-booted VMs for instant student session allocation (<500ms cold-to-terminal), supporting 50+ concurrent active sessions on bare metal • Hardened platform security using Firecracker Jailer with chroot isolation, seccomp filters, and per-VM iptables rules preventing inter-VM traffic and unauthorized internet access • Developed Nexus Engine (Go), the orchestration backend exposing REST APIs for VM lifecycle control, session tracking, health monitoring, and billing across the entire platform • Open-sourced the platform as Nexus OSS, expanding from a single backend into a multi-component system with Engine, CLI, Node Agent, and Installer with automated release pipelines	
Cybersecurity Intern – ZecurX Cybersecurity Pvt. Ltd	Aug 2025 – Feb 2026
• Core contributor to internal CTF and cybersecurity training platform supporting web, system, and reverse-engineering labs • Built and maintained secure backend challenge infrastructure with deployment, isolation, and lifecycle management for 500+ users • Implemented automation-driven provisioning workflows aligned with DevOps infrastructure practices • Collaborated with security engineers to harden systems, validate vulnerabilities, and ensure platform reliability	

Projects

Nexus OSS – Infrastructure Platform

Go, Rust, Bash, GitLab CI

- Built a multi-component infrastructure platform: API server and control plane, Node Agent, CLI tooling, and one-line bootstrap installer
- Designed REST API with session lifecycle management, API key authentication (constant-time comparison), rate limiting, and path traversal protection
- Engineered CI/CD release pipeline with GitLab CI: multi-arch binary builds, checksum verification, generic package registry distribution, and channel-based installer (stable/dev/tagged)
- Implemented observability and operational tooling including health monitoring, structured logging, and remote debugging workflows across distributed components

Aider – Cross-Platform Security Audit Tool

Python, Bash, Linux, macOS

- Engineered CLI tool automating system misconfiguration and vulnerability audits across platforms
- Reduced manual audit time by 75% with AI-based detection achieving 90% prediction accuracy
- Secured 1st place in national technical paper presentation for modular design and innovation

PyBackURLs – Historical URL Recon Tool

Python, OSINT

- Built OSINT automation tool leveraging Wayback Machine, Common Crawl, and VirusTotal APIs
- Implemented smart filtering and multi-format exports for bug bounty and security research workflows

Skills

Languages: Python, Bash, Go, C, C++, Java, SQL

Platforms & Tools: Linux, Docker, Kubernetes, K3s, GitLab CI, GitHub Actions, PostgreSQL, Redis, Nginx, Prometheus, Grafana

Domains: Platform Engineering, Cloud Infrastructure, DevOps, Linux Systems, Infrastructure Automation, Containers, CI/CD, Observability, Distributed Systems, API Design, Virtualization

Operating Systems: RHEL, Fedora, Debian, Arch, macOS

Certifications

- Google Cybersecurity Professional Certificate (Coursera) – Mar 2024
- Cisco Network Security (Coursera) – Jan 2025
- Ethical Hacking Essentials (EC-Council) – Dec 2024
- IBM Linux Commands and Shell Scripting – 2024
- NVIDIA Introduction to Networking – Oct 2025

Awards & Recognition

- **1st Place – IBM National Technical Paper Presentation (2025)** – For Aider security auditing tool
- **Top 3 – YenIoT Competition (2025)** – Flipper Zero alternative prototype
- **Certificate of Appreciation (2026)** – Responsible disclosure of P1 vulnerability in university system
- **Cybersecurity Club Member** – Yenepoya University (2024-2026)